# An Empirical Analysis of ChatGPT's Impact on Developer Productivity Using Italy's ChatGPT Ban as a Natural Experiment

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## Introduction

Substantial anecdotal evidence suggests that AI-assisted coding tools such as ChatGPT [10] have significantly increased developer productivity. However, recent empirical research has shown mixed results [4, 11, 13]. While some studies find notable efficiency gains from integrating generative AI into software development workflows—for instance, faster code completion and reduced debugging time—other work points to limited or highly context-dependent effects, with improvements varying by task complexity, developer experience, and integration depth.

Despite this growing literature, much of the existing evidence is correlational or based on controlled experiments detached from real-world settings. We still know little about how sudden changes in access to such tools affect developers in practice. The short-lived ChatGPT ban in Italy in early 2023 [6] offers a rare quasi-experimental opportunity to observe developers' behavioral responses to an abrupt disruption in access to a key AI tool.

By analyzing GitHub repositories, Kreitmeir and Raschky [8, 9] documented a 50% decline in release events in Italian open source projects during the first days of the ban. However, the effects on the granular, day-to-day development process remain unquantified. It remains unclear whether developers wrote less code or if code quality declined, which could potentially lead to downstream effects. This is a critical question, as several studies now link AI assistance to negative quality outcomes, such as an increased likelihood of introducing security vulnerabilities [5, 12].

Building on Kreitmeir and Raschky's insights, we investigate additional effects the ban has had on developers by focusing on **code-level** and **issue-level metrics**. While release events capture aggregate project milestones, they provide limited insight into the underlying development dynamics. **Code-level metrics**—specifically lines of code added and deleted per developer per day—offer a direct, granular measure of coding activity that reveals whether productivity changes manifest at the level of actual code production. These metrics are particularly valuable because they capture the immediate, day-to-day impact of AI tool availability on developers' ability to generate and modify code, independent of project management decisions about when to release.

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Similarly, **issue-level metrics**—particularly bug-related issue creation—serve as an important quality indicator that complements volume measures. If AI tools help developers write code more quickly but with more defects, or conversely, if their absence leads to more bugs due to reduced assistance, this would be reflected in post-development issue tracking. Analyzing bug-related issues provides insight into whether changes in AI tool access affect not just the quantity of code produced, but also its quality and maintainability. This is especially relevant given emerging concerns about AI-assisted code quality [5, 12].

To investigate these questions, we examine how the ChatGPT ban affected developers' coding activity and code quality indicators. Specifically, we address the following research questions:

- **RQ1**: Did the ChatGPT ban cause a reduction in the daily volume of code contributed by developers in Italy, as measured through lines of code added and deleted per user per day?
- **RQ2**: Did projects affected by the ban experience an increase in bug-related issue creation in the subsequent weeks compared to the control group, indicating that the temporary loss of ChatGPT led to higher defect rates or increased post-release debugging effort?

These research questions allow us to examine both the immediate behavioral response (coding activity) and potential downstream quality impacts (bug reports) of the ban, providing a more comprehensive understanding of how the disruption in AI tool access affected the day-to-day development process.

## Related Work

### AI-Assisted Coding Tools and Developer Productivity

A growing body of empirical research has investigated the impact of AI-assisted coding tools on developer productivity. Early studies on GitHub Copilot found significant productivity gains, with developers completing tasks faster when using AI code completion [13]. More recent large-scale field experiments have provided mixed evidence: Cui et al. [4] conducted randomized controlled trials across three companies and found a 26% increase in completed tasks among developers using AI tools, with less experienced developers showing greater gains. However, Paradis et al. [11] reported more context-dependent effects in enterprise settings, suggesting that productivity improvements vary substantially by task complexity and developer experience. Becker et al. [3] examined open-source developers and found nuanced effects that depend on the type of development work being performed.

While these studies provide valuable insights, they primarily rely on controlled experiments or observational data that may not capture real-world behavioral responses to sudden changes in tool availability. Most existing work focuses on productivity gains during tool adoption, leaving open the question of how developers adapt when access to these tools is abruptly disrupted.

### Code Quality and Security Concerns

Alongside productivity gains, researchers have raised concerns about the quality of AI-generated code. Pearce et al. [12] conducted an early security assessment of GitHub Copilot and found that it frequently generates code with security vulnerabilities. More recently, Fu et al. [5] conducted an empirical study of Copilot-generated code in GitHub projects and documented a concerning prevalence of security weaknesses. These findings suggest that while AI tools may increase coding speed, they may also introduce quality trade-offs that manifest through increased bug reports and security issues.

### The ChatGPT Ban as a Natural Experiment

Kreitmeir and Raschky [9] were the first to leverage Italy's ChatGPT ban as a quasi-experimental setting to study the real-world impact of AI tool access. Using a difference-in-differences design, they analyzed GitHub release events and documented a 50% decline in releases from Italian open-source projects during the first days of the ban. This work established the ban as a valuable natural experiment and demonstrated the feasibility of using GitHub activity data to study developer behavior at scale.

However, their analysis focused on aggregate project-level outcomes (release events), which capture only high-level project milestones. The granular, day-to-day development process—including actual code production and quality indicators—remains unexamined. Our work extends this foundation by investigating code-level and issue-level metrics that provide more direct measures of developer activity and code quality.

### Measuring Developer Productivity

Prior research has employed various metrics to measure developer productivity, ranging from task completion times to lines of code contributed. Code-level metrics such as lines of code added and deleted have been widely used in empirical software engineering research as indicators of coding activity [4, 11, 13]. While imperfect proxies for productivity, these metrics capture granular development behavior that aggregate measures like releases cannot. Issue tracking systems, particularly bug reports, serve as important quality indicators in software development and have been used to assess the relationship between development practices and code quality [5].

## Methodology

### Data

The empirical analysis uses historical GitHub activity data from users in Italy, Austria, and France for the weeks surrounding the early April 2023 ban. We leverage the GH Archive dataset, which provides a comprehensive record of public GitHub events [1]. Since GH Archive data does not include developers' location information, we infer user countries based on their public profiles, following the methods from prior research [9].

### Control Group Selection

We follow Kreitmeir and Raschky [9] in selecting Austria and France as the control group for our difference-in-differences analysis. This choice is motivated by several factors that strengthen the validity of the parallel trends assumption, which is critical for causal inference in DiD designs [2]. First, Austria and France share cultural and linguistic similarities with Italy, being neighboring European countries with comparable levels of economic development and technology adoption. Second, their geographic proximity means they are likely subject to similar regional trends in software development practices, AI tool adoption, and broader technological developments. Third, these countries were not affected by the ChatGPT ban, making them suitable counterfactuals for what would have happened in Italy in the absence of the ban. This selection helps ensure that any observed differences between Italy and the control countries during the ban period are attributable to the ban itself rather than to pre-existing differences in development trends.

## Research Design

We employ a **Difference-in-Differences (DiD)** quasi-experimental design to ensure causal inference, replicating the framework from the original paper. The analysis compares the change in developer metrics in Italy (treatment group) against those in Austria and France (control group).

The ban was implemented on April 1st, 2023. Our analysis focuses on a time window spanning four weeks before the ban (March 4–31, 2023) and two weeks after the ban (April 1–14, 2023). This time window allows us to capture both the immediate impact of the ban and potential adaptation effects. Additionally, following Kreitmeir and Raschky [9], we examine the first four working days after the ban (April 3–6, 2023) separately to assess the immediate impact before circumvention behavior (such as VPN usage, as documented in the appendix) may have mitigated the ban's effects over time. This design allows us to address the research questions (RQ1 and RQ2) by examining whether the ban caused statistically significant changes in code volume and bug-related issue creation.

In contrast to the original study—which measured aggregate productivity through project releases—our analysis zooms in on code-level and issue-level metrics that capture the day-to-day development process. The main dependent variables are:

- **Code Volume**: The number of lines of code (LOC) added and deleted per developer per day. This provides a granular measure of active coding behavior.
- **Bug-Related Issue Creation**: As an indirect quality proxy, we analyze whether the number of bug-related issues (identified through issue titles and labels containing keywords such as "bug," "fix," or "error") increased in the weeks following the ban within affected repositories.

All variables will be standardized at the developer-day level to account for individual differences in coding activity. Control variables will include developer experience (tenure on GitHub), project size, and temporal trends (day-of-week effects).

## Results

### Code Volume Analysis

We begin by examining the impact of the ChatGPT ban on developers' coding activity, as measured by lines of code (LOC) added and deleted per developer per day. Table 1 presents the difference-in-differences regression results for both the two-week post-ban period and the first four working days after the ban implementation.

The coefficient on "Italy × Post-Ban" represents the treatment effect of the ban on coding activity in Italy relative to the control countries. For lines of code **added**, we find statistically significant negative effects in both time periods. Over the two-week post-ban period, developers in Italy added approximately 6,220 fewer lines of code per day compared to the control group, an effect that is statistically significant at the 5% level ($p < 0.05$). The effect is even more pronounced in the immediate aftermath: during the first four working days after the ban, the reduction in LOC added increases to approximately 8,530 lines per day, significant at the 10% level ($p < 0.10$).

This pattern suggests that the ban had an immediate and substantial impact on developers' ability or willingness to write new code, with the effect being strongest in the first few days following the ban implementation. The larger magnitude in the four-weekday period compared to the two-week period is consistent with potential adaptation or circumvention behavior over time, as developers may have found alternative ways to access ChatGPT or adjusted their workflows.

Table 1. Difference-in-Differences Estimates: Lines of Code Changes After ChatGPT Ban

| | LOC Added (2 Weeks) | LOC Added (4 Weekdays) | LOC Deleted (2 Weeks) | LOC Deleted (4 Weekdays) |
|---|---|---|---|---|
| Italy | -1880.806 | -2079.757 | -2148.903 | -2198.754 |
| | (5419.410) | (5481.783) | (1680.966) | (1613.464) |
| Post-Ban | 1877.242 | 5491.749 | -764.124 | 645.753 |
| | (2030.811) | (3461.779) | (1381.723) | (1949.744) |
| Italy × Post-Ban | -6219.835** | -8530.137* | 865.282 | -366.177 |
| | (3095.401) | (4401.253) | (2287.457) | (2559.878) |
| Constant | 18283.484*** | 19361.889*** | 7296.271*** | 6775.601*** |
| | (4242.394) | (4379.965) | (1327.804) | (1338.330) |
| Observations | 224776 | 185999 | 224776 | 185999 |
| $R^2$ | 0.000 | 0.000 | 0.000 | 0.000 |
| Adjusted $R^2$ | 0.000 | 0.000 | -0.000 | 0.000 |
| Residual Std. Error | 395751.579 (df=224766) | 401965.443 (df=185989) | 184244.846 (df=224766) | 189388.650 (df=185989) |
| F Statistic | 1.456 (df=9; 224766) | 1.709* (df=9; 185989) | 0.471 (df=9; 224766) | 1.095 (df=9; 185989) |

*Note:* *p<0.1; **p<0.05; ***p<0.01

In contrast, we find no statistically significant effects on lines of code **deleted** in either time period. The coefficients are small in magnitude and not statistically significant, suggesting that the ban did not substantially affect developers' code deletion behavior. This asymmetry—significant effects on code addition but not deletion—indicates that the ban primarily impacted developers' ability to generate new code rather than their code maintenance or refactoring activities.

These results provide initial evidence addressing RQ1, indicating that the ChatGPT ban caused a statistically significant reduction in the daily volume of code contributed by developers in Italy, specifically through reduced code addition rather than changes in code deletion patterns.

*Ban Lift Analysis.* To further examine the causal relationship, we analyze the period following the ban's lift on April 28th, 2023. Table 2 presents difference-in-differences estimates comparing coding activity before and after the ban was lifted. The coefficient on "Italy × Post-Lift" represents whether developers in Italy experienced a rebound in coding activity relative to the control countries once ChatGPT access was restored.

We find no statistically significant effects in any of the specifications. The coefficients on "Italy × Post-Lift" are small in magnitude and not statistically significant for both LOC added and deleted, across both the two-week and four-weekday time windows. This null result suggests that either (1) developers had already adapted their workflows during the ban period, (2) the reduction in coding activity persisted despite the ban being lifted, or (3) any recovery effects were gradual and not captured in the immediate post-lift period. Additionally, as shown in Appendix A, there was likely substantial ban circumvention through VPN usage and other methods, which may explain why we observe an effect after the ban was announced but not after it was lifted: by the time the ban was lifted, many developers had already circumvented the restriction and regained access to ChatGPT. The absence of a rebound effect provides additional context for interpreting the ban's impact, indicating that the disruption had lasting effects that did not immediately reverse upon restoration of access.

Table 2. Difference-in-Differences Estimates: Lines of Code Changes After ChatGPT Ban Lift

| | LOC Added (2 Weeks) | LOC Added (4 Weekdays) | LOC Deleted (2 Weeks) | LOC Deleted (4 Weekdays) |
|---|---|---|---|---|
| Italy | -4086.519 | -4010.950 | -1077.617 | -1231.823 |
| | (5620.952) | (5770.782) | (1337.319) | (1315.548) |
| Post-Lift | -1055.302 | 566.091 | 625.877 | 1660.161 |
| | (1685.454) | (2932.394) | (1173.611) | (1880.063) |
| Italy × Post-Lift | 643.460 | -2767.974 | 258.638 | -904.240 |
| | (2735.654) | (3758.932) | (2029.785) | (2714.440) |
| Constant | 16726.655*** | 17113.333*** | 6773.386*** | 7076.681*** |
| | (4418.021) | (4613.066) | (894.855) | (929.972) |
| Observations | 209142 | 169574 | 209142 | 169574 |
| $R^2$ | 0.000 | 0.000 | 0.000 | 0.000 |
| Adjusted $R^2$ | 0.000 | 0.000 | -0.000 | -0.000 |
| Residual Std. Error | 378708.112 (df=209132) | 390728.730 (df=169564) | 190160.556 (df=209132) | 197850.531 (df=169564) |
| F Statistic | 0.616 (df=9; 209132) | 0.691 (df=9; 169564) | 0.714 (df=9; 209132) | 1.561 (df=9; 169564) |
| *Note:* | | | | *p<0.1; **p<0.05; ***p<0.01 |

## Code Quality Indicators

[Analysis of bug-related issue creation to be completed]

## Discussion

## Conclusion

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## A Ban Circumvention Analysis

This appendix presents the difference-in-differences analysis of search volume for circumvention tools—specifically VPN services and Tor Browser—in Italy (treatment group) compared to Austria and Germany (control groups) before and after the ban implementation on April 1st, 2023. The data was obtained from Google Trends [7], which provides normalized search interest values representing the share of web searches for a given term relative to all searches on Google over time. The values are normalized on a scale of 0–100, where 100 represents peak popularity for the term during the selected time period.

Figure 1 shows the VPN search volume over time for Italy, Austria, and Germany. The vertical dashed line indicates the ban start date (April 1st, 2023). Prior to the ban, Italy showed consistently lower baseline search interest in VPNs compared to the control countries, with average values around 21 points versus 33–35 points for Austria and Germany. Following the ban implementation, Italy experienced a dramatic spike in VPN search interest, reaching a peak of 100 (the maximum normalized value) on April 1st, and maintaining elevated levels throughout the post-ban period with an average of 51 points.

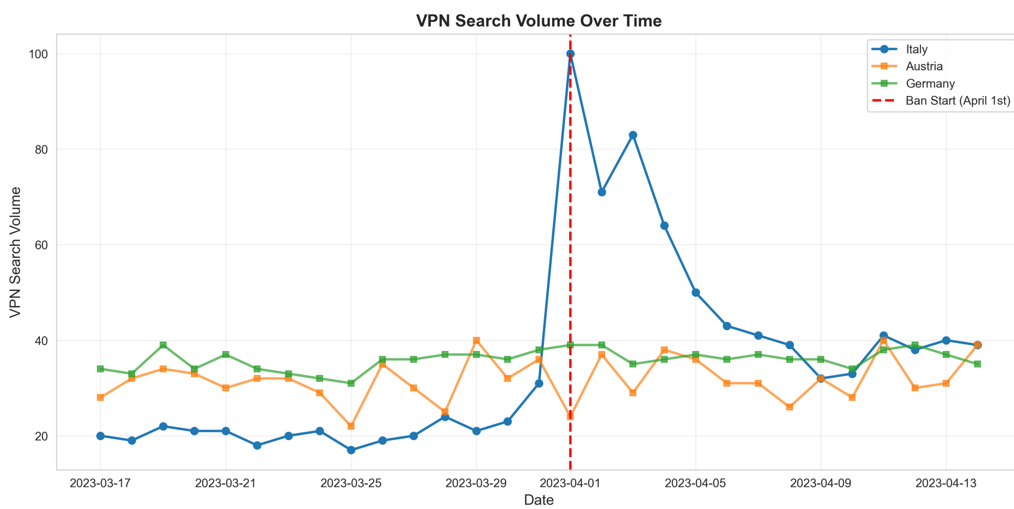


Fig. 1. VPN Search Volume Over Time

Figure 2 compares the average VPN search volume before and after the ban implementation across the three countries. The visualization clearly demonstrates the substantial increase in VPN search interest in Italy following the ban, while the control countries (Austria and Germany) showed minimal changes, consistent with the parallel trends assumption required for difference-in-differences estimation.

Table 3 presents the difference-in-differences regression results for both VPN and Tor Browser searches. The dependent variable in each model is search volume (normalized Google Trends index). The coefficient on "Italy ×

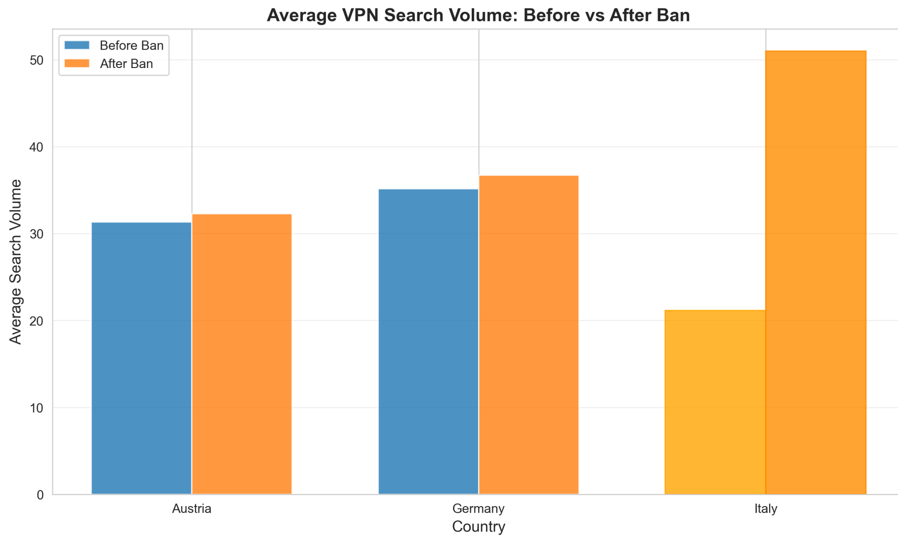


Fig. 2. Average VPN Search Volume: Before vs After Ban

Post-Ban" represents the treatment effect for each circumvention tool. For VPN searches, the coefficient indicates that searches in Italy increased by 28.6 points (on the 0–100 scale) more than in the control countries after the ban implementation, an effect that is statistically significant at the 1% level ($p < 0.001$). For Tor Browser searches, the treatment effect is 20.2 points, statistically significant at the 5% level ($p < 0.05$). These results suggest a substantial increase in interest in both VPN services and Tor Browser as means of circumventing the ban. The magnitude of the VPN effect represents more than a doubling of baseline search interest in Italy, consistent with users seeking tools to bypass geographic restrictions.

Table 3. Difference-in-Differences Regression Results

| | *Dependent variable: search_volume* | |
|---|---|---|
| | VPN | Tor Browser |
| Italy | -12.100*** | -10.167 |
| | (2.823) | (6.221) |
| Post-Ban | 1.267 | -0.710 |
| | (2.346) | (5.169) |
| Italy × Post-Ban | 28.600*** | 20.167** |
| | (4.063) | (8.953) |
| Constant | 33.233*** | 17.567*** |
| | (1.630) | (3.592) |
| Observations | 87 | 87 |
| $R^2$ | 0.497 | 0.079 |
| Adjusted $R^2$ | 0.479 | 0.046 |
| Residual Std. Error | 8.928 (df=83) | 19.672 (df=83) |
| F Statistic | 27.344*** (df=3; 83) | 2.371* (df=3; 83) |
| *Note:* | | *p<0.1; **p<0.05; ***p<0.01 |